Capacitors

We're going to build on the skills we learned in the LED circuit. This circuit uses a capacitor. A capacitor stores current and releases it at a steady rate. The current in a circuit can change with power fluctuations. Power fluctuations can damage sensitive circuits. Capacitors are used to smooth out the current supplied to components. They are also used to gradually reduce the amount of current being supplied to a component.

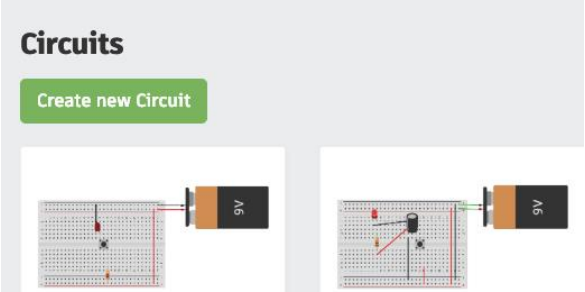
A capacitor is much like a battery. It stores current and releases that current when a circuit is closed. They don't store as much current like a battery. Capacitors are reliable and don't need to be replaced like batteries.

This is a good time to discuss potential energy and kinetic energy. Batteries and capacitors store electricity. This storage is potential energy. The stored energy is released into circuits and provides kinetic energy. The amount of kinetic energy released is measured as current. The potential energy in batteries is measured in volts.

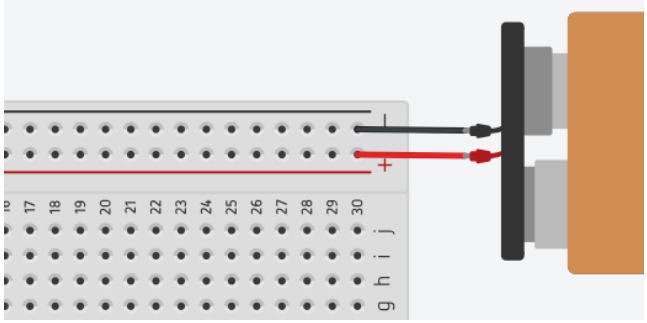
The potential energy in capacitors is measured in Farads. Capacitors store electrostatic charge. They don't produce the current released. They store it and release it at a steady rate.

A Capacitor Circuit

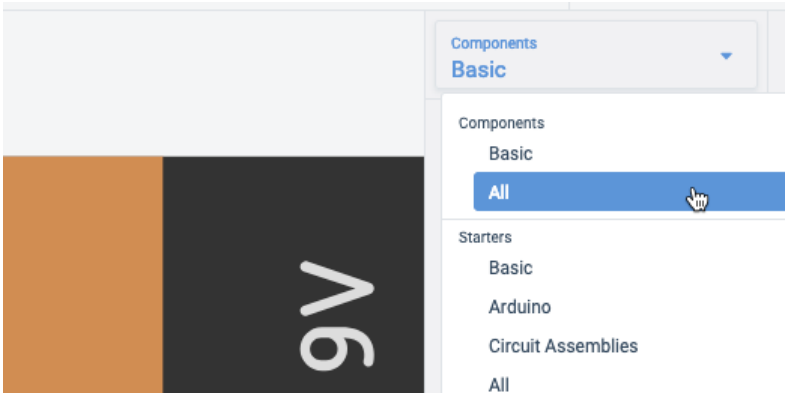
Log into your Tinkercad account. Go to the circuit development environment. Create a new circuit project.



Place a breadboard on the work area. Place a 9-volt battery next to the breadboard and connect the terminals to one end of the breadboard. Color code the jumper wires from the battery to the breadboard.



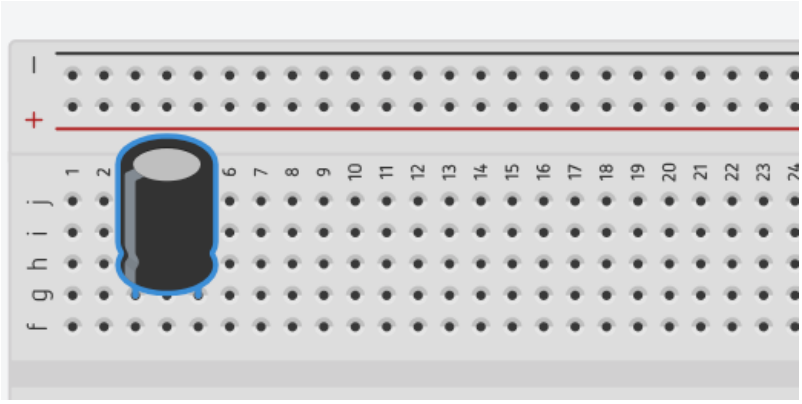
There are two types of capacitors available in the component panel. The capacitor available in the basic components panel is a ceramic capacitor. The capacitor we need is in the advanced components list. Click the component's selector and choose All.



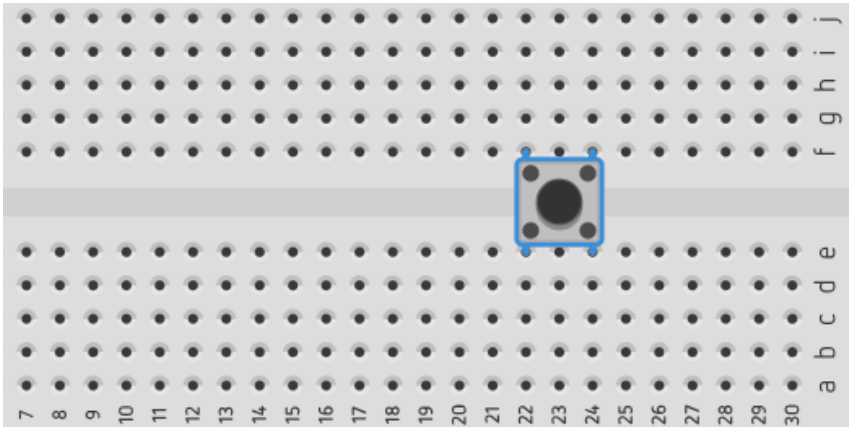
Select the polarized capacitor and take it over to the breadboard. This capacitor is a lot like a diode. It accepts current from one end of the component and passes it through to the other side.



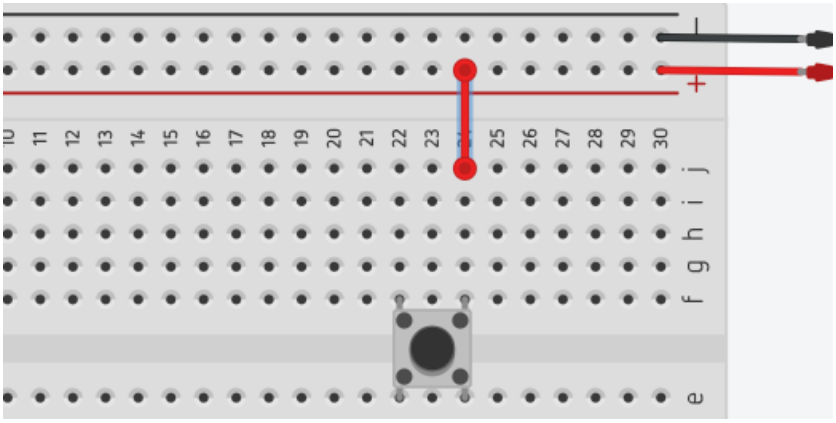
The capacitor is very large. It takes up a lot of space on the breadboard. We are going to work around it while setting up the rest of the components. Place the capacitor on the left side of the board for now.



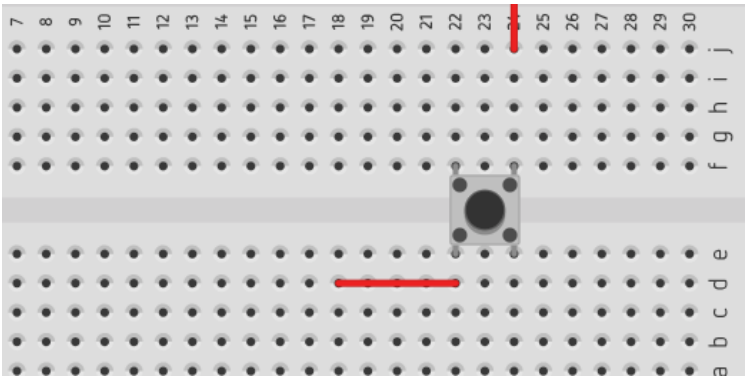
Get a push-button and place it on the right side of the board. Make sure it bridges the two halves of the breadboard.



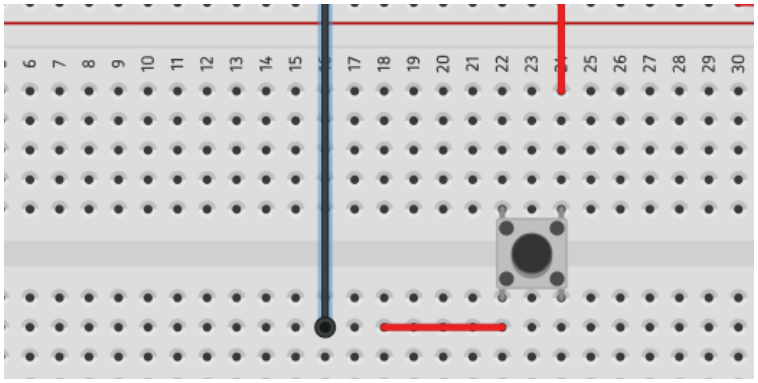
Connect a jumper wire from the positive rail to the right side of the button. Color code the jumper wire.



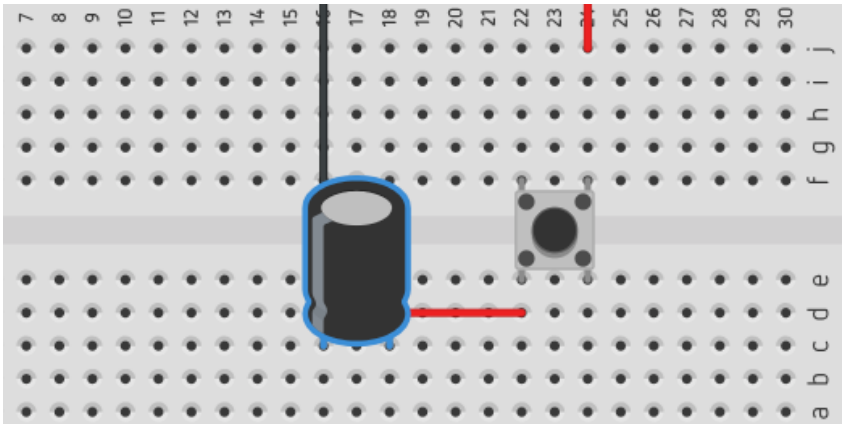
Click once on the hole below the lead on the left side of the push button. Count four holes to the left and click once on that hole. This will create a connection between the row connected to the left lead on the push button and the selected row. This wire is necessary to accommodate the large capacitor.



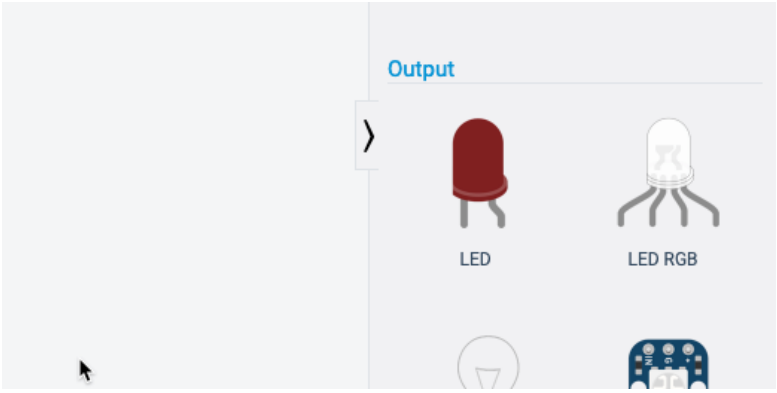
Skip a hole and click once to start a new jumper wire. Connect this wire to the negative rail. We skipped a hole to accommodate the leads on the capacitor.



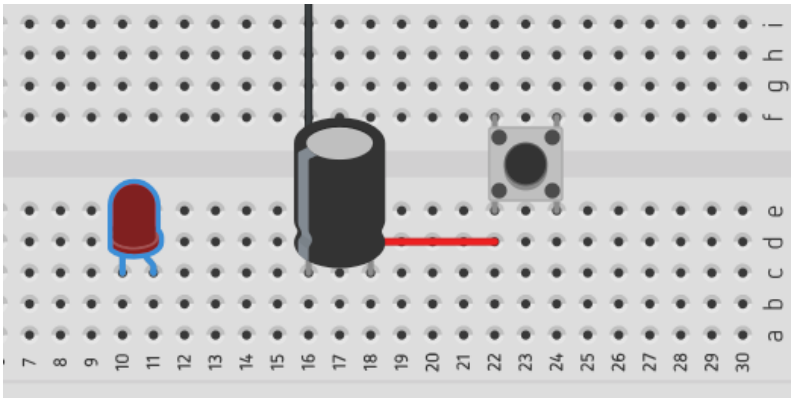
Get the capacitor from the left side of the breadboard. Place it so that the leads match the rows with our jumper wires. The jumper wires are hidden behind the capacitor.



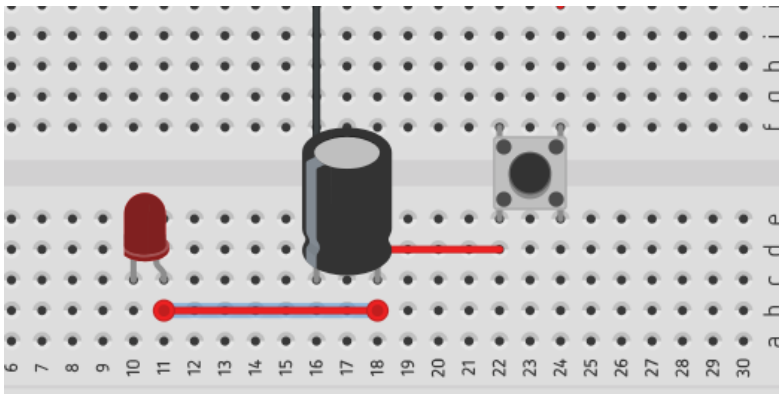
Place an LED on the breadboard. You will need to scroll down to the output section of the advanced components panel.



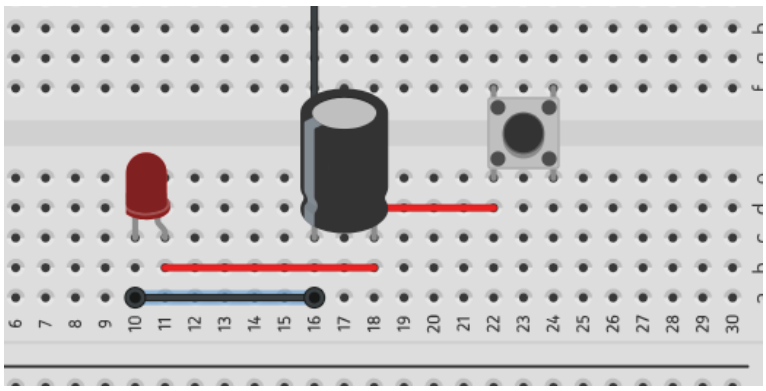
Place it a few spaces to the left of the capacitor.



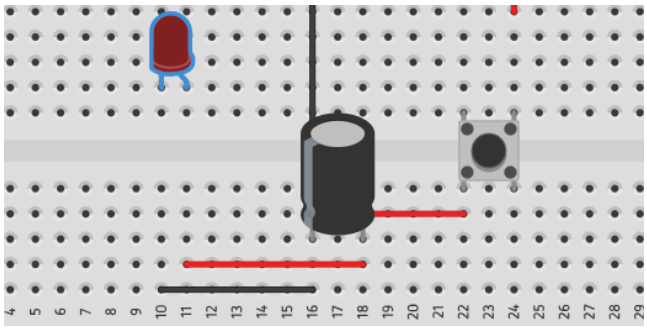
Click one of the holes in the row connected to the positive lead of the capacitor. Run the jumper wire to the row connected to the Anode on the LED.



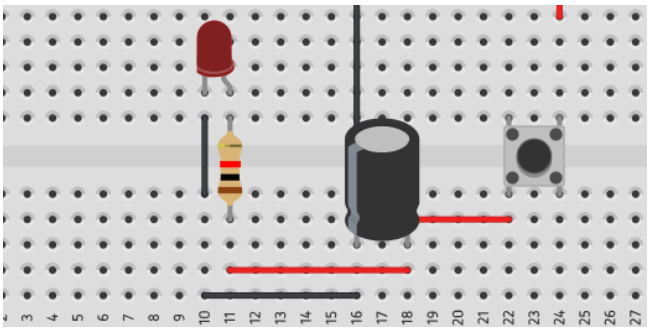
Connect another jumper wire from the row connected to the negative terminal on the capacitor and connect it to the Cathode row for the LED.



We need a resistor to prevent overloading our LED. We need some space for the resistor. Click and drag the LED to the top half of the breadboard. Leave a column empty for jumper wires.

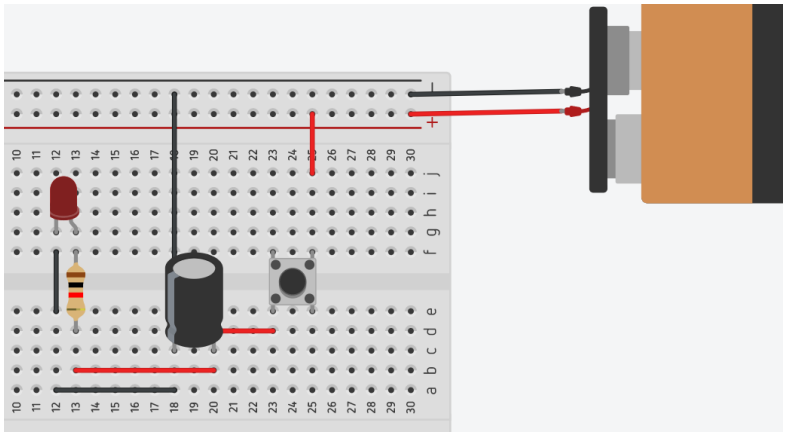


Scroll back to the top of the advanced components panel. Get the resistor and connect it to the Anode lead on the LED. Place a jumper wire connection between the lower half of the breadboard and the Cathode row on the LED. Look at the image below for the completed circuit.

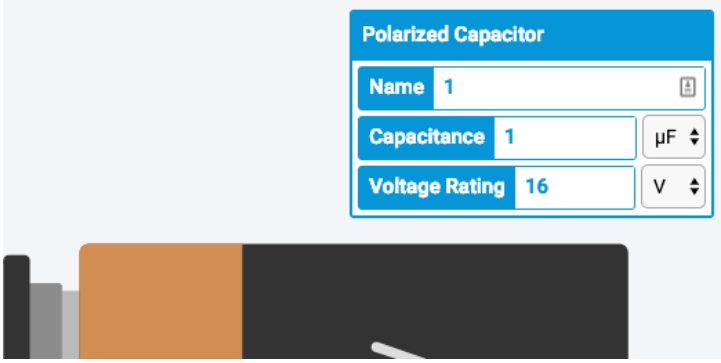


The completed circuit

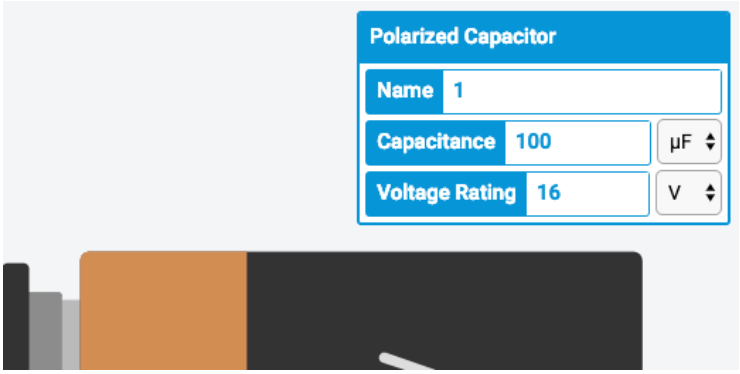
Press the Start Simulation button and press the push button. The LED lights up when we push the button. This isn't any different from our first circuit. We need to make some adjustments to our capacitor to exaggerate the effects we are producing. Stop the simulation and click once on the capacitor.



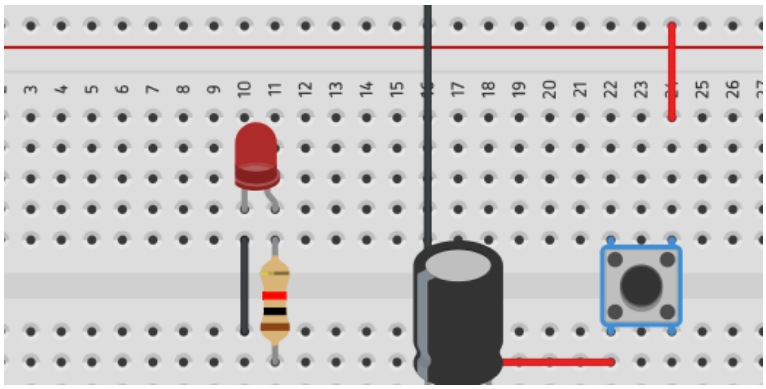
Capacitors have different levels of capacitance. This is the amount of electrostatic charge a capacitor can hold. The capacitance is measured in Farads. The capacitor is currently set to emulate a capacitor with one microfarad. The little “u” symbol next to the letter “f” represents micro.



Stop the simulation and change the value from 1 to 100 microfarads. You need to stop the simulation each time a value is changed. You can change the value while the simulator is running but the results will still be based on the original value. Stopping and starting the simulator updates the values in the simulation. Start the simulation again and press the push button.



The LED will light as before. The difference appears when we release the button. The LED dims gradually before it completely turns off. The LED is dimly light because the capacitor is still discharging current into the LED. It can’t store as much current like a battery so the LED isn’t as bright.

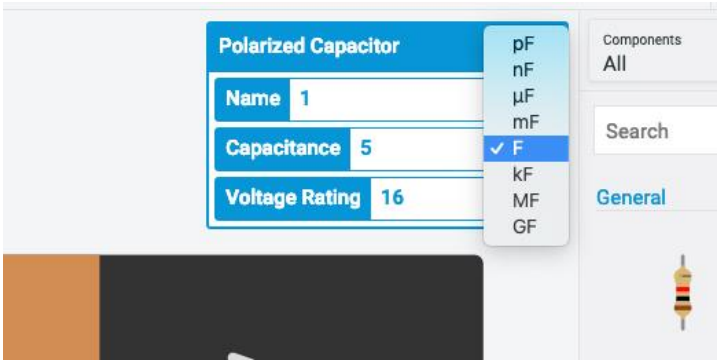


We can increase the time an LED remains light by increasing the capacity of the capacitor. To better see the effect, we need to increase the value by a lot. Change the value to 1,000. Don't forget to stop and start the simulation. Press and release the button. The LED will remain light much longer. It will take about a minute to completely turn off.

Let's increase the value one more time to see what else is going on. Click the Farads unit selector and choose Farads. There are different units of measure. They are like using inches, feet, yards, and miles. Picofarads is the smallest. Farads is the main unit of measure. Change the capacitance value to 5 Farads. Run the simulation.

The simulation shows the LED gradually going from dark to bright. The LED remains bright for much longer after we release the button.

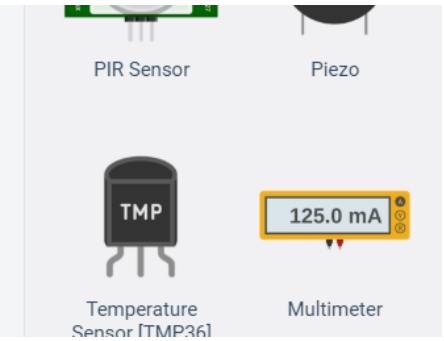
The LED gradually gets brighter because the Capacitor isn't full yet. The capacitor needs to fill up with enough electrostatic charge before it is released. Increasing the capacitance increases the electrostatic charge needing to be filled before releasing the built up charge to a component.



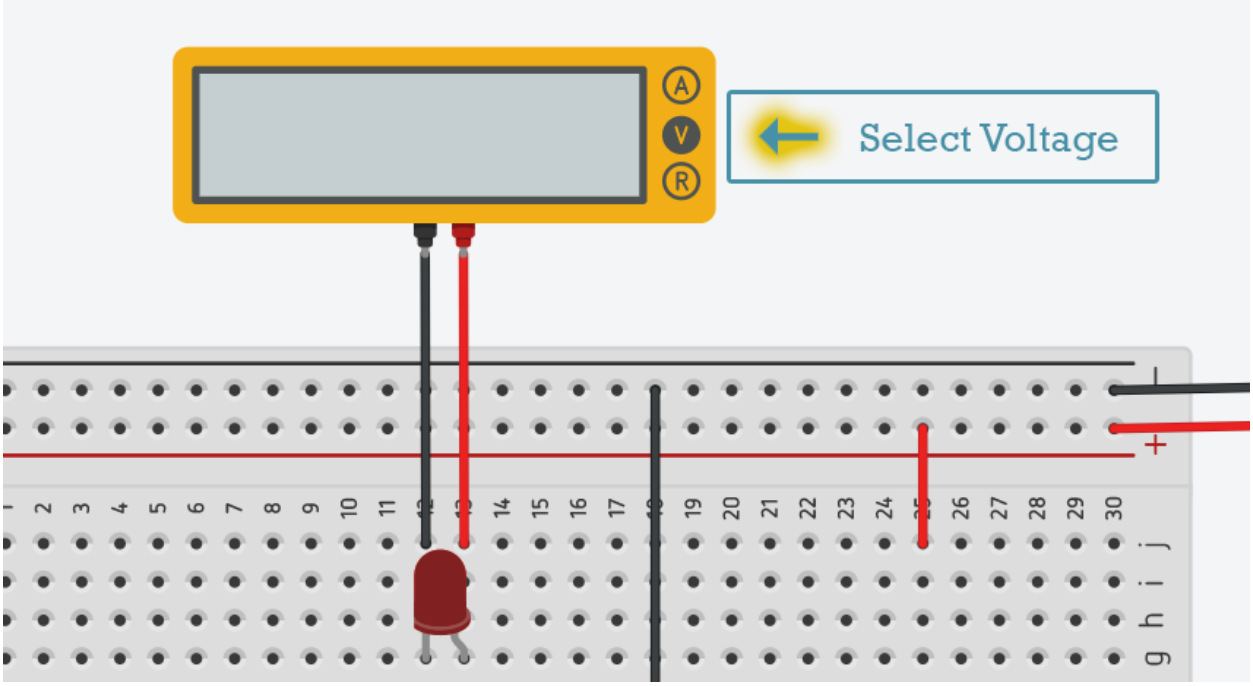
Let's take it one step further to drive the point home. Increase the value to 20. Start the simulation and press the button. Keep the button pressed. The LED doesn't light up right away this time. It takes longer to fill the capacitor before it discharges. It takes about five seconds for the LED to begin glowing.

Look at the numbers

To wrap this up, let's use a Multimeter to observe what is happening *voltage-wise* as this circuit is being simulated. Grab a Multimeter to add to the circuit:



Connect the Multimeter leads to the same breadboard rows as the LED. Make sure the select the “V” on the Multimeter so that it is set to measure the voltage on the circuit at that point.



Begin the simulation. At this point the Multimeter should read 0.00V. Once you press the button we’ll be able to see the forward voltage that reaches the LED – this is impacted by the **Battery Voltage** as well as the **resistor** in the circuit.

When we release the button, however, we can see charge continued to be supplied by the **Capacitor.** The values and rate at which the voltage drop will occur depend on the **Capacitance Value** that the resistor is set to. After voltage drops past a certain point, there won’t be enough voltage to illuminate the LED. You’ll see the voltage continues to drop after this point.

